Ulrike Romatschke <romatsch@ucar.edu>

dPSDs

2 messages

Schvartzman Cohenca, David <dschvart@ou.edu> Fri, Mar 14, 2025 at 6:08 PM
To: John Hubbert <hubbert@ucar.edu>, "Romatsch, Ulrike" <romatsch@ucar.edu>

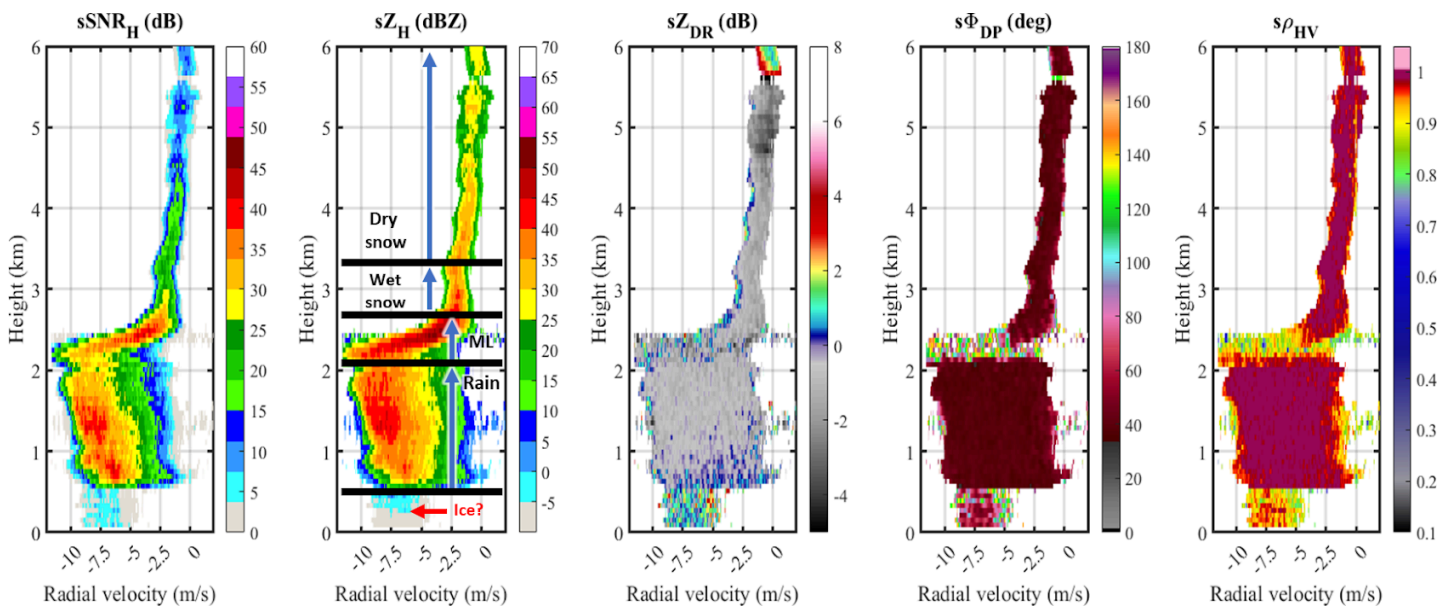
Hi John and Rike,

I'm sending my implementation of the *bootstrap* polarimetric spectral estimator, in the attached file. Run the script "`bootstrap_dpsd_example.m`" to see how it works. The bootstrap itself is in the function "`bootstrap_dpsd.m`".

Download the MATLAB scripts here: https://www.dropbox.com/scl/fi/cq9rijmncqdtrmjcudksb/Bootstrap_dPSD.zip?rlkey=1xgw5d7ixn03r1okcfd486k9m&dl=0

The technique was proposed by Umeyama et al. (2017), and is described in this paper, in IEEE TGRS: https://ieeexplore.ieee.org/document/7819424

Here's an interesting example, collected during a freezing-rain storm in Norman, with the radar in vertical pointing mode (see notes in the reflectivity panel).



And more examples (from a presentation I gave at NASA a couple years ago) at the end. See the "spectral hydrometeor classification".

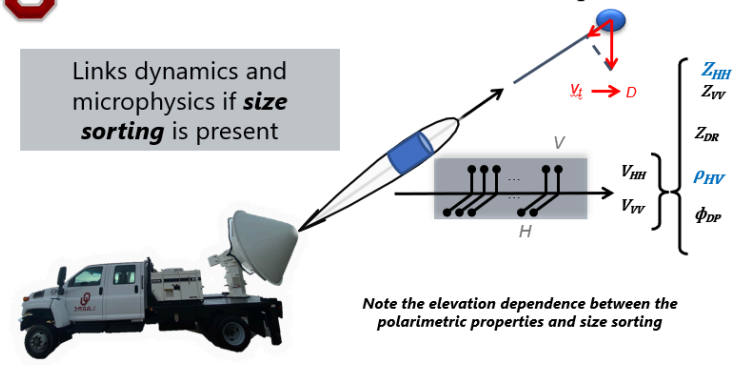
Let me know what you think, and if the code I'm sending works on your end.

Best,

-David

Polarimetric Spectral Densities

Links dynamics and microphysics if ***size sorting*** is present



Note the elevation dependence between the polarimetric properties and size sorting

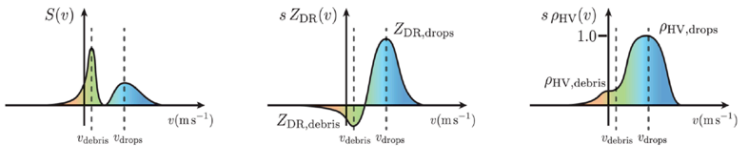
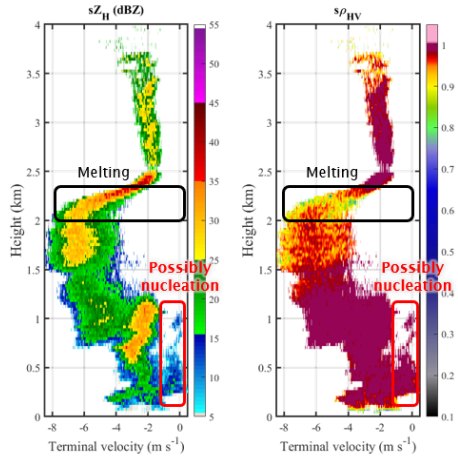
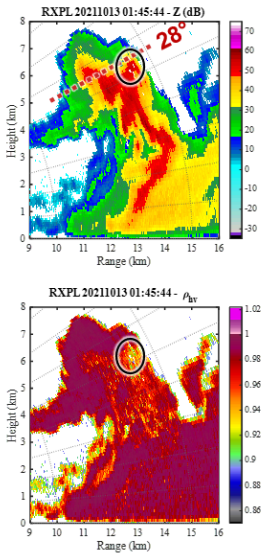


Figure 3.6: Idealized plots of (a) $s\hat{S}(v)$ (dB), (b) $s\hat{Z}_{DR}(v)$ (dB), and (c) $s\hat{\rho}_{HV}(v)$.

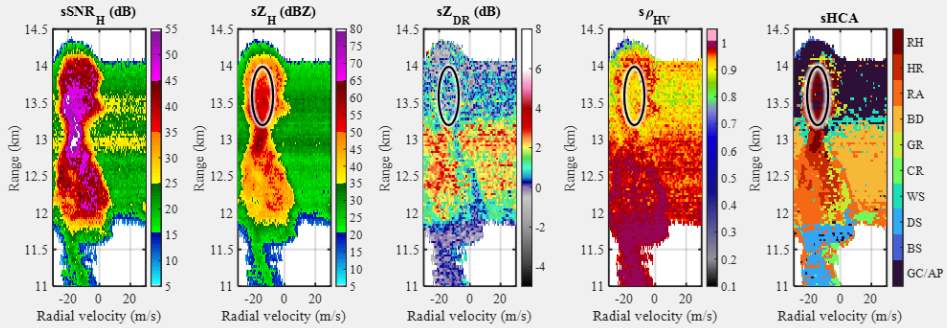


Data are from 2/25/2022 ~14:39 UTC

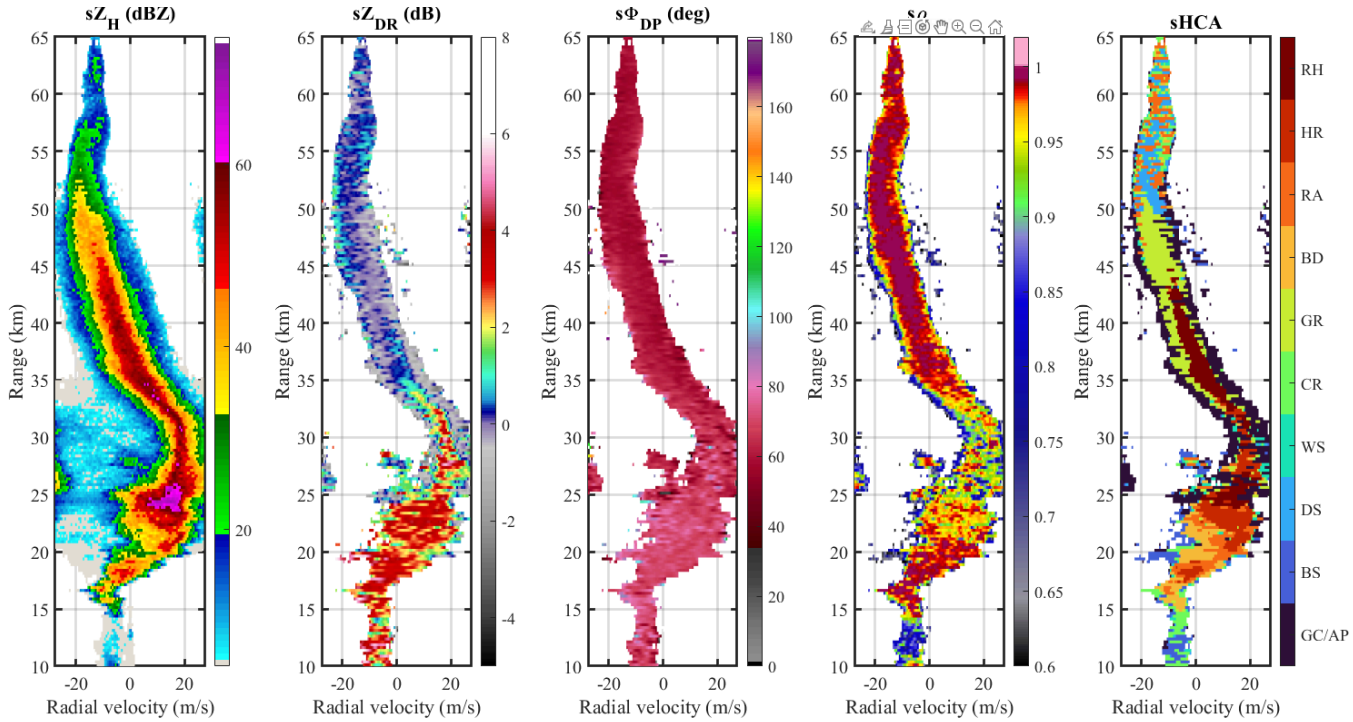
Spectral Hydrometeor Classification Algorithm (sHCA)



Elevation = 28°



These *very preliminary results* show conceptually consistent results in the spectral HCA (sHCA)



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John Hubbert <hubbert@ucar.edu> Fri, Mar 14, 2025 at 6:32 PM
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Cc: "Romatsch, Ulrike" <romatsch@ucar.edu>

..and thanks for the code and the images! -john
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